



Elementary Science

Magnets & Electromagnets

Level 4

Topic 10: Electricity & Magnets | Answer Sheet 4

Name: _____

Date: _____

A Magnetic or non-magnetic?

-  paper clip magnetic not magnetic
-  iron nail magnetic not magnetic
-  coin magnetic not magnetic
-  wood block magnetic not magnetic
-  plastic spoon magnetic not magnetic
-  steel screw magnetic not magnetic

B Poles: attract or repel?

- N S attract repel
- N N attract repel
- S S attract repel
- S N attract repel

Unlike poles attract. Like poles repel.

C Electromagnet investigation.

Same iron nail and wire; only the number of cells changed



Cells	Paper clips picked up
1 cell	3
2 cells	6

3 cells

9

What pattern do you notice?

More cells made the electromagnet stronger.

D Answer the questions.

- What is wrapped around the iron nail?
wire
- What provides electrical energy?
the cells / battery
- Which setup was strongest?
3 cells
- What happens when the current is switched off?
the electromagnet stops attracting

Magnets can push or pull — electromagnets use electricity!